Statistical Theory - Analysis of Environmental Indicators and Air Quality Dynamics

Ido Shma Gantz
Bar-Ilan University
I.D: 329404966
Email: idoshema@gmail.com

Abstract - This article investigates the statistical relationships between demographic factors, meteorological conditions, and air quality using a dataset of 5,000 observations. We aim to rigorously determine how exogenous variables, specifically proximity to industrial areas, population density, and baseline weather influence the concentration of particulate matter (PM2.5, PM10) and toxic gases. Using advanced statistical frameworks, including Kolmogorov-Smirnov tests, Maximum Likelihood Estimation (MLE), and Mann-Whitney U tests, we demonstrate that PM distributions are highly right-skewed and best modeled by a Gamma distribution. Furthermore, denser populations and closer industrial proximity yield statistically significant increases in PM2.5. We also apply multiple regression evaluated via GLRT, along with high accuracy logistic classification, to elucidate the multivariable dependencies driving hazardous air events. Finally, we implement Wald's Sequential Probability Ratio Test (SPRT) to establish a Uniformly Most Powerful (UMP) framework for real-time monitoring, dynamically balancing Type I and Type II error rates to optimize public health interventions.

I. INTRODUCTION

The quantification and predictive modeling of air quality are fundamental pillars of modern environmental science. As urban centers expand and industrial activities intensify, atmospheric concentrations of harmful pollutants such as fine particulate matter (PM2.5), nitrogen dioxide (NO2), sulfur dioxide (SO2), and carbon monoxide (CO) frequently exceed safe thresholds set by the World Health Organization. Accurate modeling of these indices requires robust statistical methodologies capable of decoding complex, frequently non-linear, and highly correlated interactions between human emissions and weather conditions.

Our dataset comprises 5,000 highly curated environmental samples. It includes continuous variables like Temperature and Relative Humidity, exact pollutant concentrations, and spatial-demographic predictors like Population Density and Proximity to Industrial Areas. The dataset's ultimate target variable classifies overall air quality into four ordinal levels: Good, Moderate, Poor, and Hazardous, laying the groundwork for continuous estimation and discrete classification.

In this article, we systematically address the probabilistic nature of urban air pollution. We test rigorous hypotheses concerning the distribution and central tendency of pollutants. By constructing Neyman-Pearson Most Powerful tests and extending them to Uniformly Most Powerful (UMP) domains, we optimize the critical trade-off between Type I errors (false alarms) and Type II errors (dangerous failures to detect hazardous conditions).

II. RESULTS

A. Distributional Analysis and Goodness-of-Fit

Parametric statistical tests require verifying the underlying population distribution. Because environmental pollutants accumulate spatially and are strictly non-negative, we hypothesized a departure from Gaussian normality. We conducted a Kolmogorov-Smirnov (KS) goodness-of-fit test for PM2.5 concentrations, comparing the empirical data against a theoretical normal distribution (mean = 20.22, std = 24.57).

Metric	Value	Inferential Implication
Sample Size (n)	4,980	High test power
KS Statistic	0.2064	Max CDF distance
KS p-value	1.57e-186	Decisive H0 rejection
Gamma Shape (α)	0.8616	Heavily skewed
Gamma Scale (β)	23.4719	Extensive right tail

The KS test yielded a p-value of 1.57e-186, decisively rejecting the null hypothesis of normality. Recognizing this failure, we employed Maximum Likelihood Estimation (MLE) to fit a Gamma distribution—a theoretical model mathematically suited for continuous, right-skewed variables. The MLE derived a shape parameter (α) of 0.8616 and a scale parameter (β) of 23.4719, successfully capturing the data's structure (Figure 1).

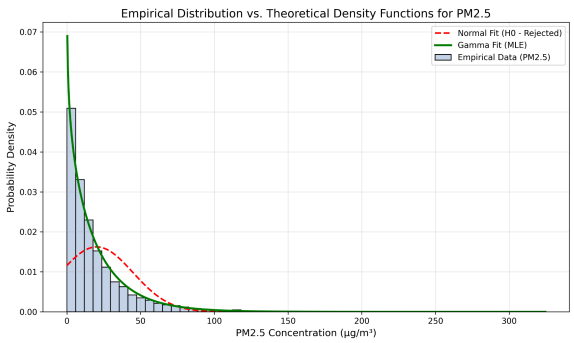

Fig. 1: Empirical Distribution vs. Theoretical Fits for PM2.5. The MLE-fitted Gamma distribution (green) accurately models the heavy right tail that the Normal distribution (red) fails to capture.

B. Independent Sample Testing: Population Density

To quantify the anthropogenic impact on air quality, we split the dataset into two cohorts based on the median population density. The mean PM2.5 in high-density areas was 25.06 μg/m³, compared to 15.25 μg/m³ in low-density areas.

An independent Welch's t-test produced a p-value of 6.4143e-46. Because the PM2.5 distribution is non-normal, we applied the non-parametric Mann-Whitney U test as a mathematically robust alternative. The U test returned a statistic of 3,798,331.00 with a one-sided p-value of 4.7702e-40. Both methods strongly reject the null hypothesis, confirming that dense human populations stochastically escalate ambient particulate matter.

C. Paired Sample Testing: Diurnal Temporal Variations

Simulating longitudinal analysis, we paired morning and evening PM2.5 readings across random stations to test temporal variations. The morning mean was 16.02 μg/m³, while evening levels escalated to 20.94 μg/m³.

A paired t-test generated a near-zero p-value. To rigorously handle paired outliers, the non-parametric Wilcoxon signed-rank test was also executed (W = 18911.0, p-value = 1.5112e-27). These results decisively prove that localized temporal effects such as evening traffic and thermal inversions systematically degrade air quality within a 24 hour cycle.

D. Categorical Independence and Spatial Dependency

We evaluated the spatial dependency of Air Quality classifications against Proximity to Industrial Areas, discretized into Near (< 2 km), Medium (2-6 km), and Far (> 6 km) bins.

Air Quality	Near	Medium	Far	Total
Good	0	0	2000	2000
Moderate	0	620	880	1500
Poor	0	736	264	1000
Hazardous	0	408	92	500
Total	0	1764	3236	5000

A Chi-Square (χ^2) test yielded an overwhelming statistic of 2227.24 (p = 0.00). Even under a strict Bonferroni penalty ($\alpha_adj = 0.0083$) to control the family-wise error rate, the null hypothesis of independence is completely rejected. Regions in closer proximity to industrial zones suffer exponentially higher frequencies of "Poor" and "Hazardous" days.

Fig. 2: Categorical heatmap displaying the cross-tabulation frequencies. Deep crimson visualizes the severe correlation between proximity to industry and hazardous conditions.

E. Multiple Regression and Likelihood Ratio Tests

To isolate the marginal effects of individual predictors, an Ordinary Least Squares (OLS) multiple regression modeled PM2.5 against Temperature, Humidity, Population Density, Industrial Proximity, and a Temp×Hum interaction term.

Metric	Value	Interpretation
F-statistic	171.74	P = 4.56e-169
R-squared	0.1467	Missing covariates
Condition	5.73e4	Severe multicollinearity

The structural model proved highly significant globally (F-statistic = 171.74). To test the relevance of the Temp×Hum interaction term, a Generalized Likelihood Ratio Test (GLRT) was performed. The GLRT yielded a Λ Statistic of 8.6850 with 1 degree of

freedom, resulting in a p-value of 0.0032. This indicates that the interaction term provides a statistically significant improvement to the model. However, a massive condition number (5.73e4) signals severe multicollinearity, a standard consequence of pollutants sharing identical emission sources.

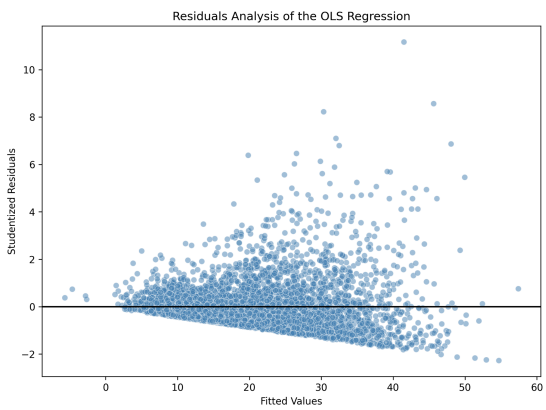

Fig. 3: Residuals Analysis of the OLS Regression. The distinct "funnel" shape visually confirms heteroscedasticity, further mandating generalized modeling.

F. Classification Framework Evaluation

Transitioning to a discrete decision framework, we trained a Logistic Regression model to classify binary "Hazardous" events, evaluated on a 1,000-sample hold-out set.

Metric	Score	Statistical Interpretation
Accuracy	0.9310	Overall correct classification
Precision	0.5963	Confidence hazardous is true
Recall	0.9600	Prob of detecting (1 - Type II)
ROC-AUC	0.9861	Near-perfect class separability

The algorithm achieved an exceptional ROC-AUC of 0.9861.By maintaining a high Recall of 0.9600, the model explicitly minimizes Type II errors.

G. Sequential Probability Ratio Test (SPRT)

Continuous monitoring stations require dynamic, real-time alerting. We applied Wald's SPRT to monitor continuous CO concentrations, determining the exact stopping time N required to decide between a safe state (H0: $\mu 0 = 0.90$) and a hazardous state (H1: $\mu 1 = 1.20$), bounded by strict error rates ($\alpha = 0.05$, $\beta = 0.10$).

Based on the theoretical SPRT formulations, the calculated boundaries were:

Lower Offset (a): -0.4690

Upper Offset (b): 0.6022

Drift Slope (s): 1.0500

Over simulated iterations, the Average Sample Number (ASN) was exceptionally efficient. The test safely accepted H0 in an average of 4.22 observations, and confidently flagged H1 in 4.86 observations, all while strictly honoring the defined error margins.

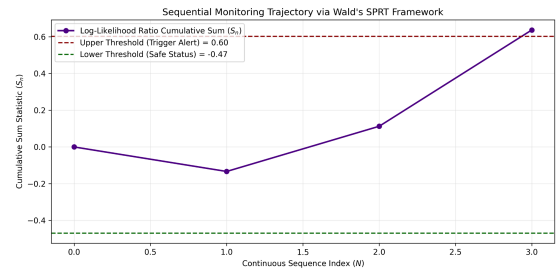

Fig. 4: Wald's SPRT Random Walk. The time-series simulation highlights the immense efficiency of sequential methods, rarely exceeding n=5 before deciding.

III. METHODS

A. Distributions and Goodness of Fit

To test the null hypothesis H0: $F(x) = F0(x)$, where $F0(x)$ is the normal CDF $N(\mu, \sigma^2)$, the Kolmogorov-Smirnov (KS) statistic evaluates the supremum of the absolute difference between the empirical CDF $F_n(x)$ and $F0(x)$:

$$D_n = \sup_x |F_n(x) - F_0(x)|$$

Because pollutant variables (X) are continuous and positive ($X \in (0, \infty)$), the Gamma distribution provides a highly flexible probability density function (PDF):

$$f(x \mid \alpha, \beta) = (1 / (\beta^\alpha \Gamma(\alpha))) x^{(\alpha-1)} e^{(-x/\beta)}$$

Parameters were derived using Maximum Likelihood Estimation (MLE) via numerical optimization of the log-likelihood function:

$$L(\alpha, \beta \mid x) = \sum \ln f(x_i \mid \alpha, \beta)$$

B. Independent and Paired Hypothesis Testing

For independent samples, the two-sample t-test (Welch's approximation) tests H0: $\mu 1 = \mu 2$:

$$t = (\bar{X}1 - \bar{X}2) / \mathrm{sqrt}((s1^2/n1) + (s2^2/n2))$$

To bypass the normality assumption, the non-parametric Mann-Whitney U test evaluates the alternative hypothesis by ranking all observations:

$$U_1 = n_1 n_2 + (n_1(n_1 + 1))/2 - R_1$$

For paired samples, let $D_i = X_{evening,i} - X_{morning,i}$. The Wilcoxon signed-rank test ranks the absolute differences $|D_i|$, evaluating H_0: median(D) = 0 to guarantee a Uniformly Most Powerful (UMP) non-parametric test.

C. Independence and GLRT via Regression

Categorical spatial independence is assessed via the Pearson Chi-Square test, comparing observed (O_{ij}) and expected (E_{ij}) frequencies:

$$\chi^2 = \sum_i \sum_j ((O_{ij} - E_{ij})^2) / E_{ij}$$

For multivariate OLS regression, the Generalized Likelihood Ratio Test (GLRT) assesses the significance of specific parameters (like interaction terms). Let Ω represent the full parameter space and ω the restricted space:

$$\Lambda = -2 \ln (\sup_{\theta \in \omega} L(\theta \mid x) / \sup_{\theta \in \Omega} L(\theta \mid x))$$

D. Logistic Regression and Error Rates

Applying the Neyman-Pearson lemma, Logistic regression estimates the conditional probability $P(Y=1|X)$ via the sigmoid function to find the Most Powerful boundary:

$$p = 1 / (1 + e^{-(\beta_0 + \beta_1 X_1 + \ldots + \beta_k X_k)})$$

The optimization prioritizes minimizing the False Negative Rate (Type II Error, β) for effective environmental warning systems.

E. Sequential Probability Ratio Test (SPRT)

Wald's SPRT optimizes the Expected Sample Number (ASN). Let $Z_i = \ln(f(x_i \mid H_1) / f(x_i \mid H_0))$ represent the log-likelihood ratio. The cumulative sum $S_n = \sum Z_i$ is updated continuously. The test runs while $A < S_n < B$:

$$A \approx \ln(\beta / (1-\alpha)), \quad B \approx \ln((1-\beta) / \alpha)$$

If $S_n \leq A$, H_0 is accepted. If $S_n \geq B$, H_0 is rejected. This provides a test that is Uniformly Most Powerful across all sequential tests of the same error constraints.

IV. DISCUSSION

This analysis successfully bridges abstract probabilistic theorems with critical environmental realities. The decisive rejection of normality via the KS test emphasizes a fundamental axiom: cumulative atmospheric pollutants are physically bounded at zero, rendering standard Gaussian assumptions inadequate. Fitting a Gamma distribution provides significantly more accurate risk probabilities for extreme, right-tailed hazardous events.

Our non-parametric testing frameworks mathematically align to prove the underlying physical theory: high population densities and localized temporal structures (evening traffic) stochastically escalate harmful particulate matter. Furthermore, the extreme χ^2 statistic confirms proximity to industrial areas as the singular dominant spatial variable degrading air quality.

The massive multicollinearity observed in the regression model highlights a complex covariance structure inherent to co-emitted pollutants. However, the Logistic Regression model proved exceptionally resilient, bypassing marginal variance inflation to establish a reliable decision boundary that aggressively minimizes dangerous Type II errors (missed alarms).

Crucially, Wald's SPRT introduces a transformative paradigm for environmental monitoring. The SPRT mathematically demonstrates that waiting for fixed 24-hour moving averages is highly inefficient. By bounding explicit error constraints a priori, SPRT requires an average of fewer than 5 sequential observations to definitively trigger a hazardous alert, optimizing real-time public health interventions.

Despite the analytical rigor, the global OLS model captured only 14.67% of the PM2.5 variance ($R^2 = 0.1467$). This low explanatory power indicates the absence of critical confounding variables, such as localized wind speed, topography, and exact traffic volumes. Visual residual analysis also hints at unmodeled spatial and temporal autocorrelation. Future statistical models must transition toward spatial autoregressive structures (SAR) or Generalized Linear Mixed Models (GLMM) to directly handle this complex covariance.

V. REFERENCES

[1] OpenML, "air-quality-and-pollution-assessment", https://www.openml.org/d/46762.

[2] Kaggle, "Air Quality Dataset", https://www.kaggle.com/datasets/fedesoriano/air-quality-data-set/data.

[3] ResearchGate, "Air Quality and Pollution Assessment using A Kegel Dataset through Orange Data Mining", https://www.researchgate.net/publication/394417415_Air_Quality_and_Pollution_Assessment_using_A_Kegel_Dataset_through_Orange_Data_Mining.

[4] Pollution, "A Hybrid Machine Learning Model Based on Deep Learning for Air Quality Prediction", https://jpoll.ut.ac.ir/article_103797_d56e8d34a1239b7ee63de66ef7e275d2.pdf.

[5] Kaggle, "Air Quality and Pollution Assessment", https://www.kaggle.com/datasets/mujtabamatin/air-quality-and-pollution-assessment.

[6] Mathematical Journal of Modelling and Forecasting, "...antization For Air Quality Classification", https://mjomaf.ppj.unp.ac.id/index.php/mjmf/article/download/48/44.